Numerical Methods Assignment-2025

Deadline – Dec 22, 2025

1. Implementation of Gauss Elimination method using Python.
2. Implement the following methods using Python:
    - Gauss Jordan
    - Gauss Jacobi
    - Gauss Seidel
3. Write a Python program to implement Jacobi's method for finding the Eigen Values, Eigen Vectors of A, real symmetric matrix.
4. Implement Spline, B Spline and Catmull Rom Spline using Python.
5. Implement the following interpolation methods and numerical differentiation using Python: -
    - Lagrange's Interpolation
    - Divided difference interpolation
    - Forward difference interpolation
    - Backward difference interpolation
    - Numerical differentiation using the above implemented interpolation

6. Implement the following numerical integration rules using Python:-
    - Trapezoidal rule
    - Simpson's rule
    - Composite Trapezoidal rule
    - Composite Simpson's rule